

TEAU 1055 – Math for Automotive Technicians (2 Credits)

Course Description

Mathematical skills form the foundation of automotive design and operation. Math for Automotive Technicians, is designed to provide realistic problems technicians will face including: whole numbers, decimals, fractions, geometry, angle measurement, metric system, integers, ratios, business statistic, repair orders, automotive engine systems, automotive electrical systems, automotive drivetrain, automotive chassis, and automotive HVAC systems.

Course Objectives

- Calculate differential gear ratios, engine bore and stroke, electrical current flow, voltage, resistance and many other components and systems found in the modern automobile.
- Determine weights of gasses needed in the air conditioning systems and fuel systems.
- Determine fuel composition, and diagnose and repair vehicles.

Course Outline

- Whole Numbers and Decimals
- Fractions and Geometry
- Integers, Signed Numbers, Ratios, Proportions, and Percent
- Business, Statistics, and Repair Orders
- The Automobile Engine and Systems
- Automobile Electrical Systems
- The Automobile Drive Train
- The Automobile Chassis
- Automobile Heating, Ventilation, and Air Conditioning
- Measurement Tools

Textbook & Reading Materials

MindTap Applied Math, 2 terms (12 months) Instant Access for Peterson/deKryger's Math for the Automotive Trade, 6th ed. (ISBN: 9781337621854)

Assignments & Assessments

Student Contract
Orientation
Orientation Acknowledgement
Remind Txt Group
Automotive Student OE Instructions
Cengage Unlimited Student Dashboard
Cengage Technical Support
Webex Classroom Link
Calculator
Cell Phone
Module 1 Test Out: Whole Numbers and Decimals
Chapter 1: Whole Numbers
Adding Whole Numbers
Read
Watch
Do
Assess: Quiz 1-1
Subtracting Whole Numbers
Read
Watch
Do
Assess: Quiz 1-2
Multiplying Whole Numbers
Read
Watch
Do
Assess: Quiz 1-3
Dividing Whole Numbers
Read
Watch
Do
Assess: Quiz 1-4
Math Matters
Show What You Know - Chapter 1
Chapter 1 Review
Chapter 1 Test
Chapter 2: Decimals
Adding Decimals
Read
Watch
Do
Assess: Quiz 2-1
Subtracting Decimals
Read
Watch
Do
Assess: Quiz 2-2

Multiplying Decimals
Read
Watch
Do
Assess: Quiz 2-3
Dividing Decimals
Read
Watch
Do
Assess: Quiz 2-4
Math Matters
Show What You Know - Chapter 2
Chapter 2 Review
Chapter 2 Test
Module 1 Completion
Module 1 End
Module 2 Test Out: Fractions, Geometry, Angle
Measurement,
and the Metric System
Chapter 3: Fractions
Working with Fractions
Read
Watch
Do
Adding Fractions
Read
Watch
Do
Subtracting Fractions
Read
Watch
Do
Multiplying Fractions
Read
Watch
Do
Dividing Fractions
Read
Watch
Do
Math Matters
Assess: Quiz 3-1
Assess: Quiz 3-2
Assess: Quiz 3-3
Assess: Quiz 3-4
Assess: Quiz 3-5
Show What You Know - Chapter 3

Chapter 3 Review
Chapter 3 Test
Chapter 4: Geometry, Angle Measurement, and the Metric System
Angles
Circular Measures
Read
Watch
Do
Assess: Quiz 4-1
Circular Measures
Read
Watch
Do
Assess: Quiz 4-2
The Metric System
Read
Watch
Do
Assess: Quiz 4-3
Math Matters
Show What You Know - Chapter 4
Chapter 4 Review
Chapter 4 Test
Module 2 Completion
Module 2 End
Module 3 Test Out: Integers, Signed Numbers, Ratios, Proportions, and Percentages
Assess: Quiz 5-1
Subtracting Signed Numbers
Read
Watch
Do
Assess: Quiz 5-2
Show What You Know - Chapter 5
Math Matters
Chapter 5 Review
Chapter 5 Test
Chapter 6: Ratios, Proportions, and Percentages
Ratios
Read
Watch
Do
Assess: Quiz 6-1
Proportions
Read
Watch
Do
Assess: Quiz 6-2

Percentages
Read
Watch
Do
Assess: Quiz 6-3
Math Matters
Show What You Know - Chapter 6
Chapter 6 Review
Chapter 6 Test
Module 3 Completion
Module 3 End
Chapter 7: Business and Statistics
Discounts
Read
Watch
Do
Profit, Loss, and Commissions
Read
Watch
Do
Assess: Quiz 7-1
Interest and Payments
Read
Watch
Do
Assess: Quiz 7-2
Taxes
Read
Watch
Do
Assess: Quiz 7-3
Averages
Read
Watch
Do
Assess: Quiz 7-4
Math Matters
Show What You Know - Chapter 7
Chapter 7 Review
Chapter 7 Test
Chapter 8: Completing Repair Orders
Repair Orders
Read
Do
Apply: Chapter 8 Repair Order
Assess: Quiz 8-1
Math Matters
Show What You Know - Chapter 8
Chapter 8 Review

Chapter 8 Test
Module 4 Completion
Module 4 End
Chapter 9: The Automobile Engine
The Automobile Engine
Read
Do
Apply: Chapter 9 Repair Order
Assess: Quiz 9-1
Math Matters
Show What You Know - Chapter 9
Chapter 9 Review
Chapter 9 Test
Chapter 10: Automobile Engine Systems
Automobile Engine Systems
Read
Do
Apply: Chapter 10 Repair Order
Assess: Quiz 10-1
Math Matters
Show What You Know - Chapter 10
Chapter 10 Review
Chapter 10 Test
Module 5 Completion
Module 5 End
Chapter 11: Automobile Electrical Systems
Automobile Electrical Systems
Read
Do
Apply: Chapter 11 Repair Order
Assess: Quiz 11-1
Math Matters
Show What You Know - Chapter 11
Chapter 11 Review
Chapter 11 Test
Module 6 Completion
Module 6 End
Chapter 12: The Automobile Drive Train
The Automobile Drive Train
Read
Do
Apply: Chapter 12 Repair Order
Assess: Quiz 12-1
Math Matters
Show What You Know - Chapter 12
Chapter 12 Review
Chapter 12 Test
Module 7 Completion
Module 7 End

Chapter 13: The Automobile Chassis
The Automobile Chassis
Read
Do
Apply: Chapter 13 Repair Order
Assess: Quiz 13-1
Math Matters
Show What You Know - Chapter 13
Chapter 13 Test
Module 8 Completion
Module 8 End
Chapter 14: Automobile Heating, Ventilation, and Air
Conditioning
Automobile Heating, Ventilation, and Air Conditioning
Read
Do
Apply: Chapter 14 Repair Order
Assess: Quiz 14-1
Math Matters
Show What You Know - Chapter 14
Chapter 14 Review
Chapter 14 Test
Module 9 Completion
Module 9 End
Chapter 15: Measurement Tools
Digital Meters
Read
Watch
Do
Assess: Quiz 15-1
Scale Measurement
Read
Watch
Do
Assess: Quiz 15-2
Dial Indicators
Read
Watch
Do
Assess: Quiz 15-3
Micrometers
Read
Watch
Do
Assess: Quiz 15-4
Drill Sizes
Read
Do
Assess: Quiz 15-5

Digital Electronic Micrometers

Read

Watch

Do

Math Matters

Show What You Know - Chapter 15

Chapter 15 Review

Chapter 15 Test

Module 10 Completion

Module 10 End

Student Feedback

End of Course Survey

Subject to change. Please consult your Canvas course for the most current instructions and updates.

Classroom Hours

Mo, Tu, W, Th

8:00 AM - 12:00 AM

1:00 PM - 5:00 PM

Friday

8:00 AM - 12:00 AM

For a full list of course hours visit: [Course Schedule](#)

Instructor Contact Information

Cody Dawson — cdawson@stech.edu

Shad Esplin — sesplin@stech.edu

Dallin Robinson — drobinson@stech.edu

McKael Stapel — mstapel@stech.edu

Office Hours: By appointment

Email is the preferred method of communication; you will receive a response within 24 hours during regular business hours.

Canvas Information

Canvas is where course content, grades, and communication will reside for this course.

- Canvas URL: stech.instructure.com
 - For Canvas passwords or any other computer-related technical support contact Student Services.
 - For Regular Hours and Weekdays call (435) 586 - 2899
 - For After Hours and Weekends call (435) 865 - 3929 (Leave a message if no response)
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Policies

Course Grading

Course Grading: Students must achieve 80% (B-) or higher to pass graded work. Incomplete assignments must be redone to meet the required standards. Guidelines, rules, and expectations for completing assignments are provided in each course.

High School Power School Grades: Quarter student grades will be determined by student progress percentage. Faculty will use the higher percentage of either 1) quarter progress, or 2) cumulative progress for the current training plan year.

Grade Scale: The following grading scale will be used to determine a letter grade.

A : 94 - 100%	A- : 90 - 93%	B+ : 87 - 89%	B : 83 - 86%
B- : 80 - 82%	C+ : 77 - 79%	C : 73 - 76%	C- : 70 - 72%
D+ : 67 - 69%	D : 63 - 66%	D- : 60 - 62%	F : 0 - 59%

Course Policies

Class attendance is required during your scheduled time. Cell phones for many have become a distraction. When you are in class or lab we encourage you to keep your cell phones put away in a secure location. If you use ear buds we ask that you only use one so you can still hear the things going on around you. If you are using your phone for things other than school related items, instructors will ask you to put them away. Please follow the direction of your instructors. Those who have been asked to refrain from using your cell phone and fail to do so will be asked to meet with the Director of Transportation and student services will be notified. The program is designed to provide the student with as much hands-on work as possible. In the automotive industry you may be required to lift heavy objects and stand for hours at a time to complete work required. Technicians deal with chemicals and materials which require caution, these will be identified in the Right to Know Agreement provided to you. You will also be required to use computers to track and complete work.

Additional Information

InformaCast Statement: Southwest Tech uses InformaCast to ensure the safety and well-being of our students. In times of emergency, such as weather closures and delays, this app allows us to promptly deliver notifications directly to your mobile devices. To stay informed and receive real-time updates, we encourage all students to sign up for notifications. Your safety is our priority, and staying connected ensures a swift response to any unforeseen circumstances. More information and directions for signing up are available at: <https://stech.edu/emergency-notifications/>

Internet Acceptable Use Policy: The student is expected to review and follow the Southwest Technical College Internet Safety Policy at: <https://stech.edu/students/policies/>

Student Code of Conduct Policy: The student is expected to review and follow the Southwest Technical College Student Code of Conduct Policy at: <https://stech.edu/students/policies/>

Accommodations: Students with medical, psychological, learning, or other disabilities desiring accommodations or services under ADA, must contact the Student Services Office. Student Services determines eligibility for and authorizes the provision of these accommodations and services. Students must voluntarily disclose that they have a disability, request an accommodation, and provide documentation of their disability. Students with disabilities may apply for accommodations, based on an eligible disability, through the Student Services office located at 757 W. 800 S., Cedar City, UT 84720, and by phone at (435) 586-2899. No diagnostic services are currently available through Southwest Technical College.

Safety and Building Maintenance: The College has developed and follows a variety of plans to ensure the safe and effective operation of its facilities and programs. The following plans are available online: 1) Facilities Operations and Maintenance Plan; 2) Technical Infrastructure Plan; and 3) Health and Safety Plan.

Withdrawals and Refunds: Please refer to the Southwest Technical College Refund Policy at: <https://stech.edu/students/policies/>

Any high school or adult student, who declares a technical training objective is eligible for admission at Southwest Technical College (Southwest Tech). Program-specific admissions requirements may exist and will be listed on the Southwest Tech website. A high school diploma or equivalent is not required for admission but is mandatory for students seeking Title IV Federal Financial Aid.

Non-Discriminatory Policy: Southwest Technical College affirms its commitment to promote the goals of fairness and equity in all aspects of the educational enterprise, and bases its policies on the idea of global human dignity.

Southwest Tech is committed to a policy of nondiscrimination. No otherwise qualified person may be excluded from participation in or be subjected to discrimination in any course, program or activity because of race, age, color, religion, sex, pregnancy, national origin or disability. Southwest Technical College does not discriminate on the basis of sex in the education programs or activities that it operates, as required by Title IX and 34 CFR part 106. The requirement not to discriminate in education programs or activities extends to admission and employment. Inquiries about Title IX and its regulations to STECH may be referred to the Title IX Coordinator, to the Department of Education, and/or to the Office for Civil rights.

If you believe you have experienced discrimination or harassment on our campus, please contact the Title IX Coordinator, Cory Estes: cestes@stech.edu, (435) 865-3938.

For special accommodations, please contact the ADA Coordinator, Cyndie Tracy: ctracy@stech.edu, (435) 865-3944.

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